

COURSE CODE: CSC222
COURSE TITLE: DATABASE MANAGEMENT
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INTRODUCTION

The database is now such an integral part of our day-day life that often we are not aware we are using one. We consider a database to be a collection of logical related data and **DATABASE MANAGEMENT SYSTEM(DBMS)** to be the software that manages and controls access to the database.

Whenever you visit your local library there is probably a database containing details of the books in the library, details of the users, reservations and so on. There will be a computerized index which allows users to find a book based on the title, or its authors, or its subject area. The database system handle reservations to allow a user to reserve books and to be informed by post when the book is available. The system will also send out reminders to borrowers who have failed to return books on the due date. At the university, there will be a database system contained information about yourself, because you are enrolled in, details about your grant, the courses you have taken in previous years or you have taken this year and details of all your past examinations results

PURPOSE OF DATABASE SYSTEM

THE FILE PROCESSING SYSTEM (FILE BASED SYSTEM).

Consider part of a saving bank enterprise that keeps information about all customers and savings account. One way to keep your information on a computer is to store it in permanent system files. To allow users to manipulate any information, the system has a number of applications that manipulate the files, including;

A program to debit or credit an account

A program to have a new account

A program to find the balance of an account

A program to generate monthly statements.

This application programs have been written by system in response to the needs of the bank. New application programs are added to the system if the needs arises. As a result, new permanent files are created that contain infoemation about all the checking account maintained in the bank, and new application programs may need to be written to deal with situations that do not arise in saving account

The typical file processing system just described is supported by a conventional operating system permanent records are stored in various files and different application programs are written to extract records from, and to add records to the appropriate files. Before the advent of DBMSs, organizations typically store information using such system.

DEFINITION

FILE PROCESSING SYSTEM

It's a collection of application programs that performs services for end users such as the production report. Each program defines and manages its own data.

File processing system(File Based System) was an attempt to computerize the manual filing system.

LIMITATIONS OF THE FILE BASED SYSTEM

1. Separation and Isolation of Data:

When data is isolated in separate files, it is more difficult to access data that should be available.

2. Duplication of data:

Due to the decentralized approach taken by each department, a file

based system encourages uncontrolled duplication of data.

- a. Duplication is wasteful. It cost time and money to enter the data more than once. It takes up additional storage space again with associated cost. Often, the duplication of data can be avoided by sharing data files.
- b. Duplication can lead to loss of data integrity, hence the data is no longer consistent.

3. Data dependence:

The physical structure and storage of the data files and records are define in the application code. This means that changes to an existing structure is difficult.

4. Incompatibility of files:

As the structure of files in imbedded in the application programs, the structure is dependent on the application programming language. For example, the structure of the file generated by COBOL program may be different from the structure of a file generated by a C program. The direct incompatibility of such files make them difficult to process jointly. It is also costly and time consuming

5. Fixed queries/proliferation of application programs:

From the end users point of view, file based system prove to be a great improvement over manual systems. Consequently, the requirement for new modified queries grew. However, the file based system are very dependent upon the application programmers. Many queries or report that arerequired have to be written by the application programmer. As a result, two things happened. In some organization, the type of query or report that could be produced was fixed. There was no facility for asking unplanned queries either about the data itself or about which type of data are available. In others, there was a proliferation of files and application programs. Eventually, this reached a point where the work could not be handled by the data processing department current resources.

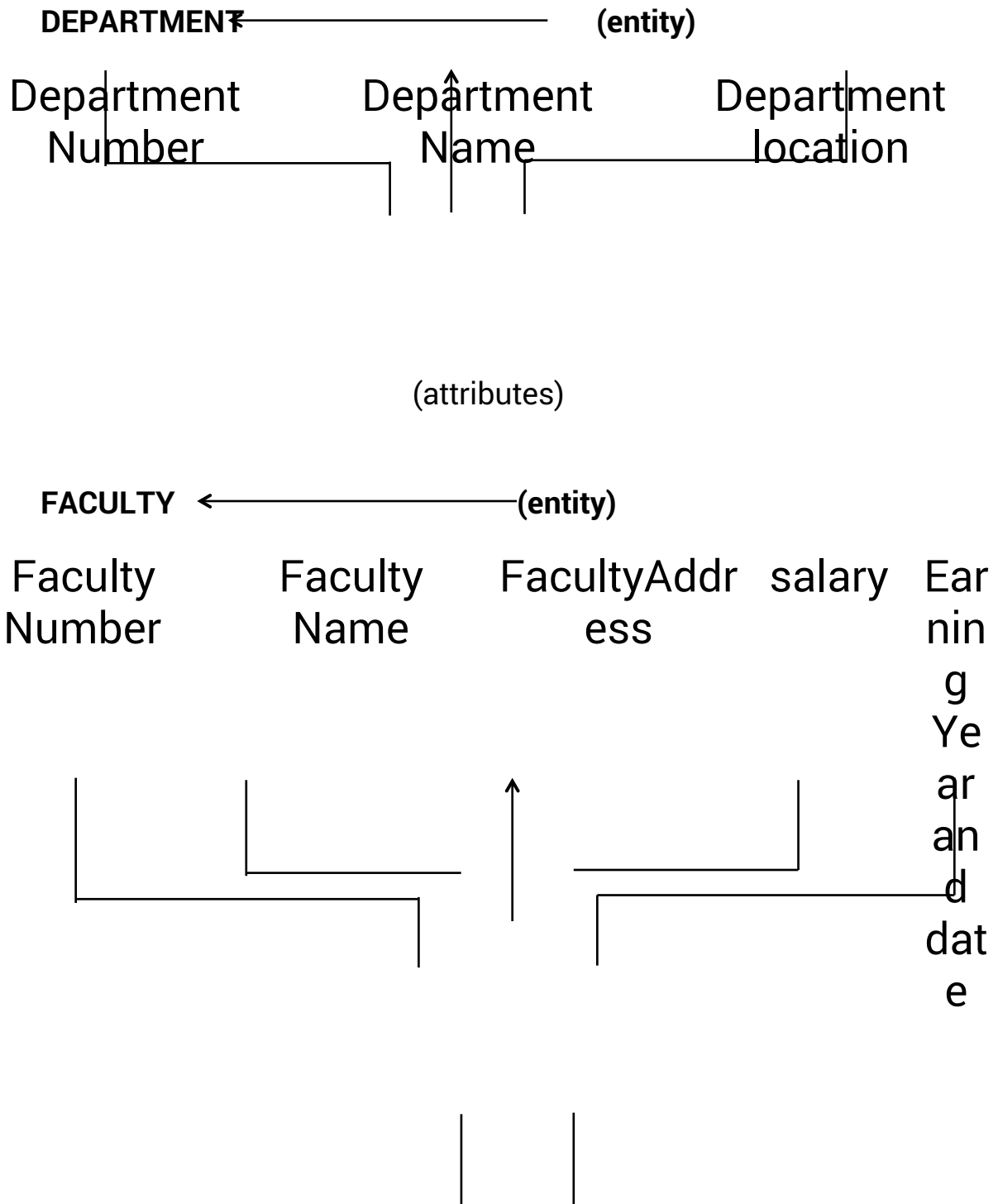
ENTITIES, ATTRIBUTE AND RELATIONSHIPS

An entity is a person, place, thing or event for which data is collected.e.g Bola, Faculty, Department, Lagos, Insurance plan etc.

An attribute is a property of an entity. Just as it is a English, the attribute of a person can include eye color and height.

At university of Ilorin, the attribute of interest for the entity called faculty members are faculty names, address etc. then this properties are not physical as eye color or height.

Figure 1 below shows two entity, department and faculty



(attributes)

FIGURE 2: ENTITIES AND ATTRIBUTES

The department entity has three attribute, department number, name and location. The faculty entity has five attribute, faculty number, name, address , salary and earnings, year to date.

A relationship is an association between entities. For example, there is a relationship between departments and faculty members. A department is related to the faculty members who work in the department, and a faculty member is related to the department in which he or she works.

Figure 2 shows the relationship between department and faculty entities.

The arrow represent the relationship, in this case, it is a one-to-many relationship. i.e one department is associated with many faculty members but each faculty member is associated with only one department.

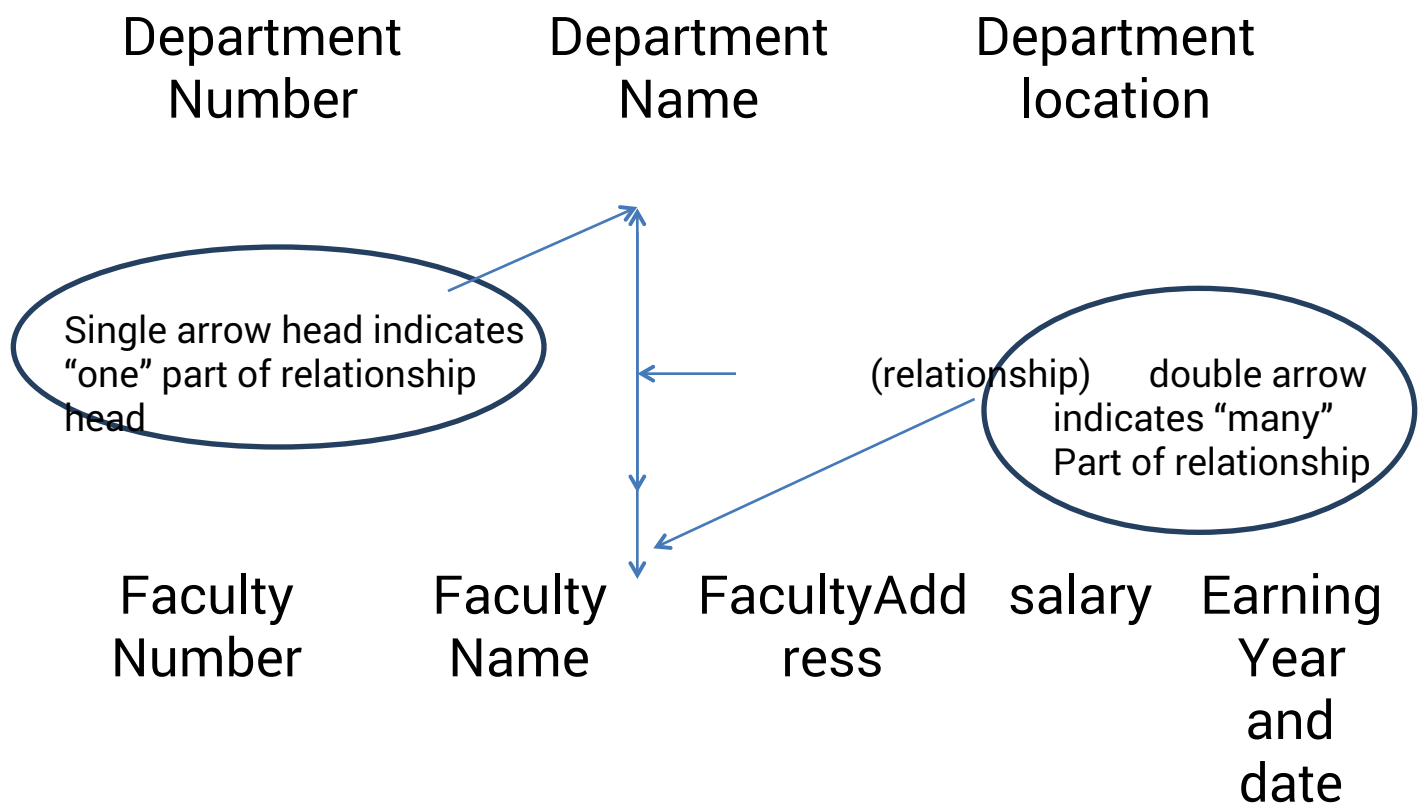


FIGURE 2: ONE-TO-MANY RELATIONSHIP

DEPARTMENT

Department
Number

Department
Name

Department
location

Single arrowhead indicate
Many parts of relationship

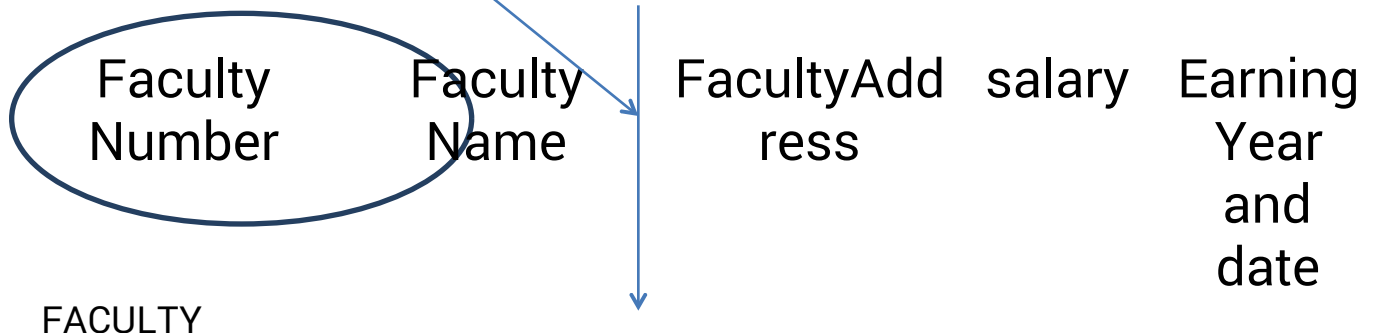


FIGURE 3

In another type of common arrangement used in a database environment. The same type of relationship is indicated by having only single arrowhead indicate the "many" part and arrow head for the "one"

The diagram in Figure 3 or the style that will be encountered most often in

this case.

FILES, RECORDS AND FIELDS

In a file processing environment, the critical terms are fields, records and file.

The smallest amount of data that can be stored in the bits are grouped into bytes (characters) .

Collections of character for a field. A field is an item of information. E.g. SURNAME, LAST NAME, SEX, AGE.

A collection of fields form and record.

Collection of occurrences of a specific type of record for a file. A file is a collection of records which contains logically related data.

DATABASES

In a typical file processing environment, each user are or unit such as payroll, personnel and COMSIT has its own collection of files and programs that access these files. See figure 4.

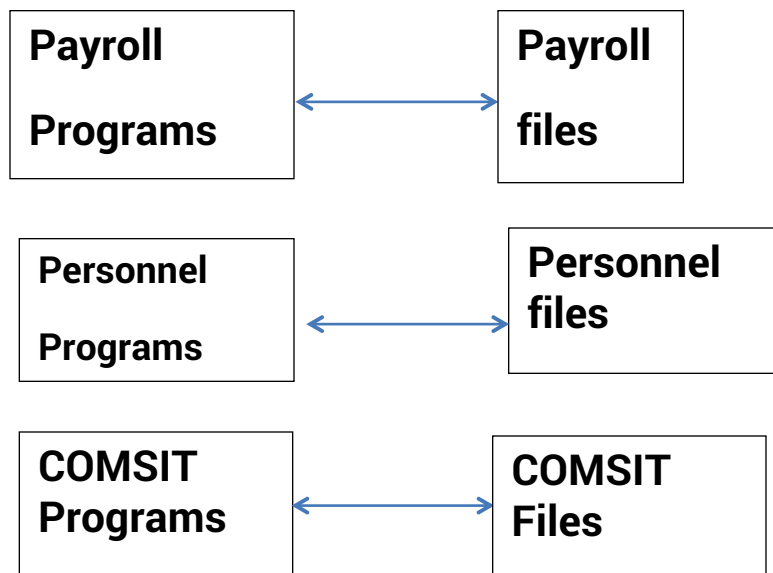


FIGURE 4: CLASSICAL FILE APPROACH

In this type of environment, there is usually overlap of data between units(user area). Hence, there is redundancy in the system. The address of a faculty member can occur in many places. This certainly wastes a lot of space and the problem it causes for update are potentially much more severe.

In addition, trying to produce report or response queries that span or cut across units can be extremely difficult. This problem lead to the idea of a pool of data, or database, rather than separate collection of individual files.

Definition: A database is a structure that can house information about multiple type of entities as well as relationships among the entities.

Note that this definition does not indicate that all the entities must be housed with the same actual physical file on disk. More than one user has access to the data stored in database.

We could eliminate the problem associated with a file processing system by moving to a database environment. One way to this would be to have a single database that users from each unit would access as shown in fig 5 below.

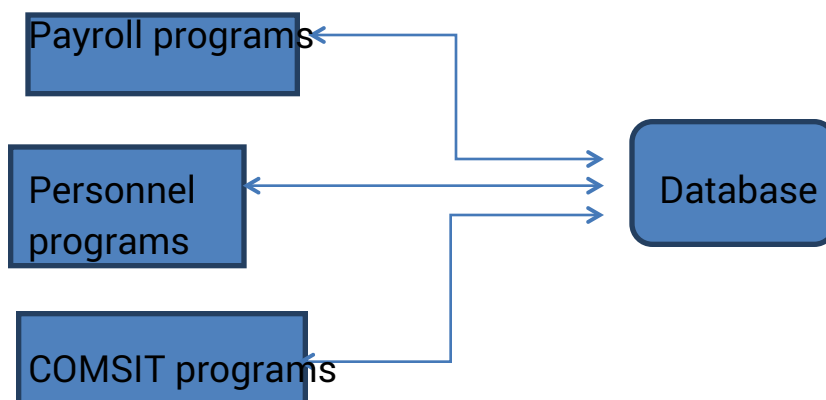


Fig 5.Database(user-maintenance)

Managing a database is a complicated task. A failure on any part of these program will destroy the structure of the database.

DATABASE MANAGEMENT SYSTEM

Software packages called **database management system**(DBMS) can be used for manipulating databases. A database management system is a software product through which users interact with a database. The actual manipulation of the underlying database structures is handle by DBMS. See the figure below

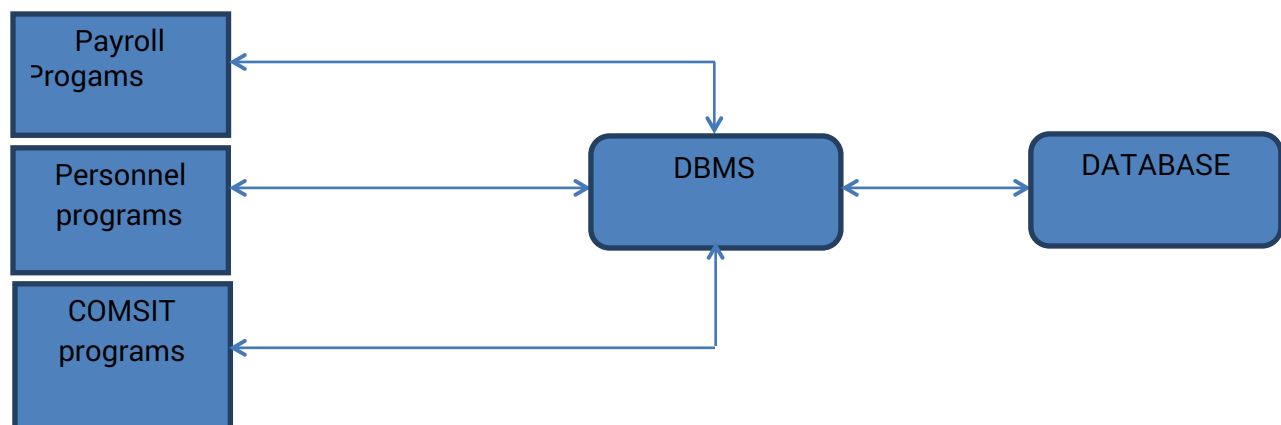


FIGURE 6: DATABASE (USING A DBMS)

VIEW OF DATA

A DBMS is a collection of interrelated files and a set of programs that allow users to access and modify these files. A major purpose of a database system is to provide users with an abstract view of the data. That is, the system hides certain details of how the data are stored and maintained.

DATA ABSTRACTION

For the system to be usable, it must retrieve data efficiently. This concern has led to the design of complex data structures for the representation of data in the database. Since many database systems users are not computer trained, developers hide the complexity from users through several levels of abstraction, to simplify users interactions with the systems. There are 3 levels of data abstract

1. Physical level
2. Logical level
3. View level.

1. Physical level:

The Lowest level of abstraction describe how the data are actually stored. At the physical level, complex low-level data structures are described in detail

2. Logical level:

The next higher level of abstraction describe what data are stored in the database and what relationships exist among those data. The entire database is thus described in terms of a small number of relatively simple structures. Although implementation of the simple structures at the

logical level may involve complex-physical-level structures. The user of logical level does not need to be aware of this complexity. The logical level of abstraction is used by database administrators, whom must decide what information is to be kept in the database.

3. View level:

The highest level of abstraction describes only part of the entire database. Despite the use of simpler structures at the logical level, some complexity remains, because of the larger size of the database. Many users of the database system will not be concerned with all this information. Instead, such users need to access only a part of the database. So that their interaction with the system is simplified. Hence the view level of abstraction is defined.

The interrelationship among these three levels of abstraction is illustrated in figure7 below

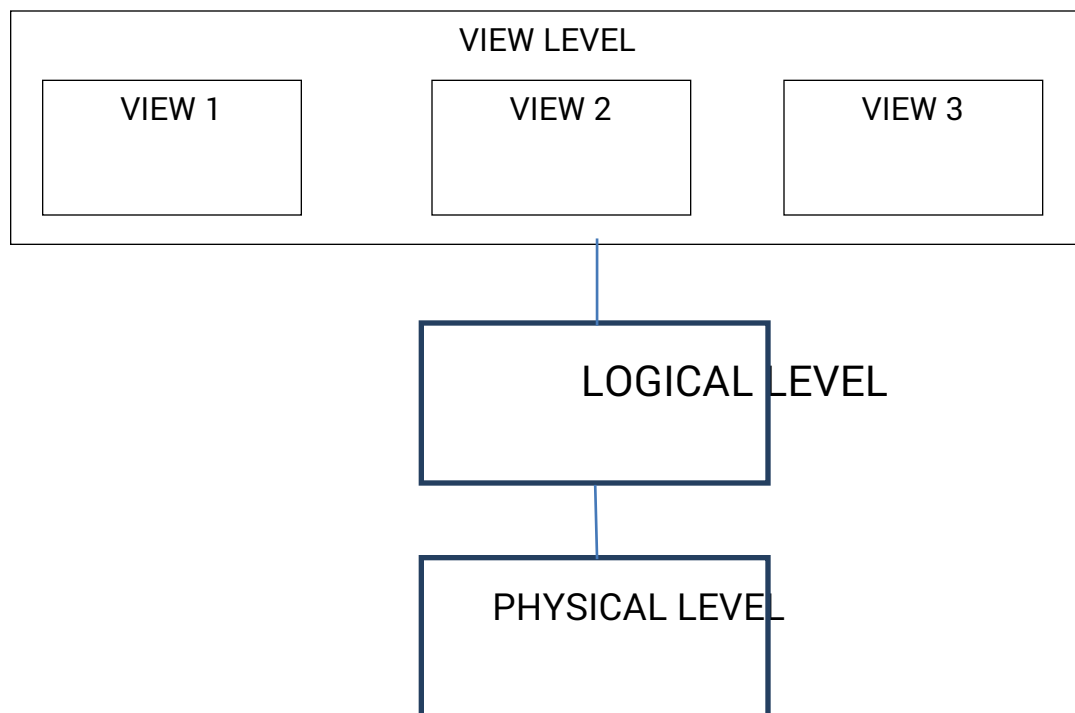


FIGURE 7: THE THREE LEVELS OF DATA ABSTRACTION

An analogy to the concept of data types in programming languages may clarify the distinction among levels of abstraction. Most high level programming languages support the notion of a record type. For example in a Pascal-like language, we may declare a record as follows.

```
Type customer = record  
Customer_name; string  
Social_security; string  
Customer_street;string  
Customer_city;string  
End;
```

This code defines a new record called customer with four fields. Each field has a name and a type associated with it. A banking enterprise may have several such record types,including:

- *account, with Fields account_number and balance
- *employee, with fields employee_nameand salary.

At the physical level, a customer, account or employee record can be described as a block of consecutive storage location (e.g words or bytes). This language compiler hides this level of detail from programmers.

INSTANCES AND SCHEMAS

Databases change over time as information is inserted and deleted. The collection of information stored in the database at a particular moment is called an instance of the database. The overall design of the database is called the database schema. Schemas are changed infrequently, analogies to the concept of data types, variable and values in programming languages is useful here.

Returning to the customer record type definition, note that, in declaring the type customer, we have not declared any variables. To declare such variables in a Pascal-like language, we write

```
Var. customer1:customer;
```

Variable customer1 now corresponds to an area of storage containing a customer type record.

A database schema corresponds to the programming language type definition. A variable of a given type has a particular value at a given instant. Thus, the value of a variable in programming language corresponds to an instance of a database schema.

Database systems have several schemas, partitioned according to the level of abstraction that have been discussed. At the lowest level is the physical schema, at the intermediate level is the logical schema, and at the highest level is a subschema. In general, database systems support one physical schema, one logical schema and several subschemas.

DATA MODELS

Database management systems are characterized by the model of data they follow. A data model has two components: structure and operations.

The structure refers to the way the system structures data or, at least the way the users of DBMS feel the data is structured. The operations are the facilities given to the users of the DBMS to manipulate data within the database. There are four models or categories for the DBMS ;

1. The relational model
2. The network model
3. The hierarchical model
4. The object-oriented model

1. The relational model: A relational model database is perceived by the user as being just a collection of tables. Formally, these tables are called relations, hence the name relational model. Relationships are implemented through common columns in two or more tables.

The figures below shows the database of each of the entity in Marvin college;faculty members,department,insurance plan, speaking topics and job history records.

FACULTY

FACULTY NUMBER	FACULTY NAME	FACULTY ADDRESS	SALARY
119	Martin, Barb	94 Ridge, Grant,MI	32,500
209	Newton, Jane	210 leonard, Ada, MI	39,000
421	Jone,Alan	121, calton, Hart, MI	30,000
462	Johnson, Jam	604, 46 th , Lowell, MI	35,000
485	Sparks, Pam	201, Olive, Hart, MI	43,500
507	Canton, Rob	5 state, cutler, MI	29,500
564	Ferris, Jim	116 W-9 th , troy, MI	38,500
683	Nels, Frank	46 crester, Ada, MI	46,500

DEPARTMENT

DEPT NO	DEPT NAME	DEPT LOCATION
1	COMPUTER SCIENCE	

JOB-HISTORY

FAC NUMB	RANK	DATEAIT
119	ASST	91581
209	ASST	90176
209	ASST	91383
421	ASST	81583
462	ASST	41271
462	ASST	91481
485	ASST	40265
485	ASST	91274
485	RAF	91383
507	ASST	91281
564	ASST	52176
564	ASST	90191
683	ASST	80274
683	PROF	81283

Consider the relational database for Marvin college. If you examine the department and faculty tables, you will see that they both contain a DEPT NUMB column. This column allows us to relate a department to the faculty members in the department. There are many ways of manipulating a relational database one of the prevalent of these is a language called SQL (structured query language) which was developed by IBM. E.g. if we wish to print the number, name and address on all faculty members whose salary is over \$35,000 we would type.

SELECT FACNUM, FACNAME, FACADDR FROM FACULTY WHERE SALARY > 35000 and the computer will respond with

FACNUM	FACNAME	FACADDR
209	Newton, Jane	210 Leonard, Ada, MI.
485	Sparks, Pam	201 Olive, Hert, MI.
564	Ferris, Tim	16 w 9th, Troy, MI.
683	Nils, Frank	46 Creston, Ada, MI.

NETWORK MODEL

A network model database is perceived by the user as a collection of record types and relationships between these record types. Such a structure is called a network. In contrast to the relational model, in which relationships are implicit (being derived from matching columns in the tables), in the network model the relationships are explicit (presented as part of the structure itself). Consider the database shown in the figure below:

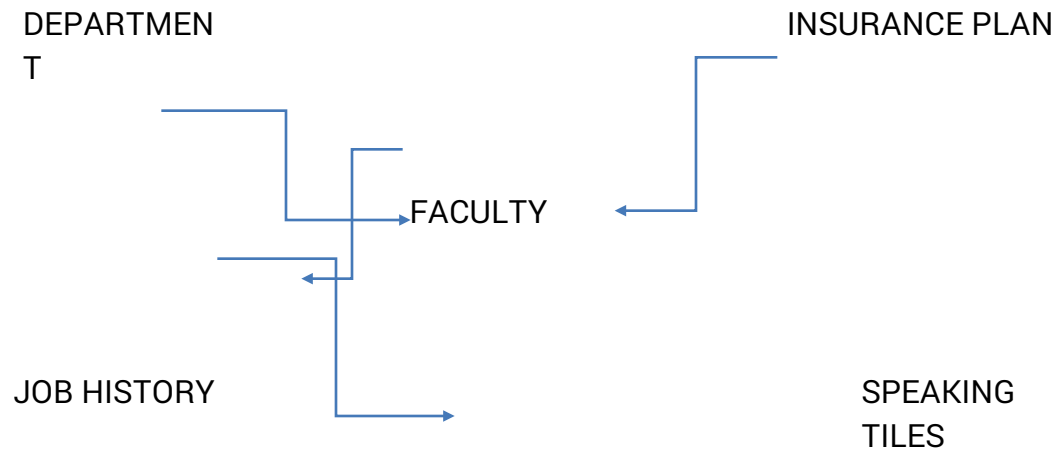


Figure 8

DEPARTMENT
 DEPTNUMB
 DEPTNAME
 DEPTLOC
 JOB-HISTORY
 BANK
 DATE ATT
 INSURANCE-PLAN
 PLANNUMB
 PLANDESC
 SPEAKING-TOPICS
 TITLE
 TIMES GIVEN
 LST GIVEN
 FACULTY
 FACNUMB
 FACNAME
 FACADDR
 SALARY
 EARN YTD
 DEGREE
 FIELD

CUR RANK
DEP CCV

The rectangles represent the record types in the database. There is a record for each of the entities: departments, faculty members, insurance plans, job history information and speaking topics. The arrows represent the relationships.

We manipulate a network model database by following the arrows. This procedure is often referred to as database navigation.

HIERACHICAL MODEL: Database is perceived by a user as a collection of hierarchies or trees. A hierarchy is really a network with an added restriction; no box can have more than one arrow entering the box.

OBJECT-ORIENTED MODEL: Database is perceived by the user as a collection of entities or objects, where each object includes the actions that can take place on that object. In the database below, we leave an object for each entity: departments, job history, insurance plan, speaking topics and faculty member. Each object can contain other objects or portion of other objects. The FACULTY object contains the INSURANCE-PLAN, DEPARTMENT and JOB-HISTORY objects.

DEPARTMENT OBJECT

DEPTNUMB
DEPTNAME
DEPLOC

JOB-HISTORY OBJECT

RANK
DATE ATT
FACULTY OBJECT, SUBSET (FAC NUM)

INSURANCE-PLAN OBJECT

PLANNUMB
PLANDESC
FACULTY OBJECT, SUBSET (FACNUMB, FACNAME, FACADDR, DEP COV).

SPEAKING-TOPICS OBJECT

TITLE
TIMESGIVEN
LST GIVEN
FACULTY OBJECT, SUBSET (FACNUM, FACNAME, FACADDR, DEGREE, FIELD)

FACULTY OBJECT

FACNUMB

FACNAME
FACADDR
SALARY
EARN YTD
DEGREE
FIELD
CURRANK
DEPCOV

INSURANCE-PLAN OBJECT
DEPARTMENT OBJECT
JOB-HISTORY OBJECT

ADVANTAGES OF DATABASE PROCESSING

1. Getting more information from same amount of data
2. Sharing of data
3. Balancing conflict requirements
4. Enforcement of standards
5. Controlled redundancy
6. Consistency
7. Integrity
8. Security
9. Data Independence

1. Getting more information from the same amount of data: The primary goal of a computer system is to turn data (recorded facts) into information. If all the data were in a common database, all authorized users will be able to access it easily.
2. Sharing of data: The data can be shared among authorized users, allowing users access to more of data. Several users will have access to the same piece of data (e.g. a faculty member's address) but use it in a variety of ways. When a faculty member's address is changed, the change is immediately available to all users.
3. Balancing conflicting requirements: For the database approach to function adequately, there must be a person or group within the organization in charge of the database itself. This group is often called Database Administrator (DBA). By keeping the overall needs of the organization in mind, DBA can structure the database to the benefit of the entire organization, not just a single user group.
4. Enforcement of standards: The central control mentioned in the previous paragraphs, DBA can ensure that standard for such things as data names, usages and formats are

followed uniformly throughout the organization.

5. **Controlled redundancy:** Since data that was kept separate in a file-oriented system is now integrated into a single database, we no longer have multiple copies of the same data.
6. **Consistency:** Since we have controlled redundancy, data in each record for various entities remain the same throughout the database.
7. **Integrity:** An integrity constraint is a rule that data in the database must follow. A database has integrity if data in the database satisfies all integrity constraints that have been established. The DBA can define validation procedures that will ensure the integrity of the database.
8. **Security:** is the prevention of access to the database by unauthorized users. Since, DBA has control over the operational data, it can define procedures to ensure that only legitimate users access the data.
9. **Data independence:** occurs when the structure of the database can change without requiring the programs that access the database to change.

DISADVANTAGES OF DATABASE PROCESSING

1. Size
 2. Complexity
 3. Cost
 4. Additional hardware requirement
 5. Higher impact of a failure
 6. Recovery more difficult
-
1. **Size:** To support all the complex functions it must provide to users, a DBMS must by its nature be a large program occupying megabytes of disk space, as well as a substantial amount of internal memory.
 2. **Complexity:** The complexity and breadth of the functions finished by a DBMS make it a complex product. Programmers and analysts must understand the features of the system to take full advantage of it.
 3. **Cost:** A good mainframe DBMS is an expensive product. By the time all the appropriate component related to the DBMS are purchased for a major mainframe system, the total

price can easily run into \$100,000 to \$400,000 such is not the case a micro-computer with cost less than \$1000.

4. Additional hardware requirements: Because of the size and complexity of a DBMS, greater hardware resources are required than would be necessary without the DBMS.
5. Higher Impact of failure: Since many of the information systems resources are now concentrated in the database, a failure of any component was a much more far reaching effect than a non-database environment.
6. Recovery more difficult: Because of the added complexity, the process of recovering the database in the event of a catastrophe is a more complicated one, particularly if the database is being updated by a large number of users concurrently (i.e. at the same time).

FUNCTIONS OF A DBMS

The main function of a DBMS is to store, update and retrieve data in a database.

1. Data storage, retrieval and update: A DBMS must provide users with the ability to store, retrieve and update data in the database.
2. A user-accessible catalogue: A DBMS must provide a catalogue in which descriptions of data items are stored and that is accessible to users.
3. Transaction Support: A DBMS must provide a mechanism that will ensure that either all the updates corresponding to a given transaction are made so that none of them is made.
4. Concurrency control services: A DBMS must provide a mechanism to ensure that the database is updated correctly when multiple users are updating the database concurrently.
5. Recovery services: A DBMS ensure the recovering of the database in the event that the database is damaged in any way.
6. Authorization services: DBMS ensures that only authorized users can access the database.
7. Support for data communication: A DBMS must be capable of integrating with communication software.
8. Integrity services: DBMS ensures that both the data in the database and changes to the data follow certain rules.
9. Services to promote data independence: A DBMS must include facilities to support the independence of programs from the actual structure of that database.
10. Utility services: A DBMS must provide a set of utility services for enhancing the proper functioning of the database.

NORMALIZATION

Normalization enables us to analyze the design of a relational database to see whether it is good or bad. In particular, normalization gives us a method for identifying the existence of potential problems called update anomalies in the design. The normalization process also supplies method for correcting these problems. The process involves various types of normal forms:

1. First normal form (1NF)
2. Second normal form (2NF) and
3. Third normal form (3NF) are three of these types. It is these three that will be of greatest use to us during database design.

They form a progression in which a relation that is in 1NF is better than a relation that is not in 1NF; a relation that is a 2NF is better yet; and so on. The goal of this process is to allow us to start with a relation or collection of relations and produce a new collection of relations equivalent to the original collection (i.e. this represent the same information but is free of problems). For practical purposes, this means that relations in the new collection will be at least in 3NF.

Two crucial concepts are fundamental to the understanding of the normalization process. They are functional dependence and keys.

DEFINITION: An attribute B, is functionally dependent on another attribute A (or possibly a collection of attributes), if a value for A determines a single value of B at any one time.

If B is functionally dependent on A we also say that A is functionally dependent on B. this is represented as follows;

$A \twoheadrightarrow B$

DEFINITION: Attribute A (or a collection of attributes) is the primary key for a relation R if:

1. All attributes in R are functionally dependent on A.
2. No sub collection of the attributes in A (assuming A is a collection of attributes and not just a single attribute) also has property 1.

OVERVIEW OF NORMALIZATION PROCESS

One of the principal objective of relational database is to ensure that each item of the data is held only once within the database. For example, if we hold customers address, the address of any one customer is represented only once through all the stages of the application.

The reason for this are first to minimize the amount of space required to hold the database, but also and more importantly to simplify maintenance of the data, if the same information is held into two or more places, then each time the data changes, each occurrence of the data must be located and amended. Also having two “copies” of the same data give rise to the possibility of being different.

In many cases, it is relatively easy to arrange the tables to meet the objective. There is however a more formal procedure called normalization that can be followed to organize data into a standard format which avoids many processing difficulties.

The purpose of normalization are:

1. To put data into a form that conforms to relational principles e.g single valued columns, each relation represent one entity
2. To avoid redundancy by storing each “fact” within the database only once.
3. To put data into a form that is more able to accommodate change.
4. To attain certain difficulties updating (so called anomalies)
5. To facilitate enforcement constraints on the desk

Normalization involves checking that the tables conform to

certain rules and if not rearranging the data. This will mean creating new tables. Normalization is a multi-stage process, the end result of each stage is called a normal form.

Successive stages produce a greater and greater degree of normalization, there are a total of seven normal forms called in increasing degree and grouped for the convenience of description.

1. First, second and third normal forms (1NF, 2NF, and 3NF)
2. Boyce-code (BCNF)
3. Fourth normal form (4NF)
4. Fifth normal form (5NF) and
5. Domain key normal form

NORMAL FORM 1ST NF, 2ND NF, AND 3RD NF

The normalization process assumes that you start with some informal description of all the data attribute that the database application appear to require, this is often called UN-NORMALIZED DATA. This section of attribute is then tested using criteria defined by each of the normalization stages. If the data fails the criteria, there is a prescribed procedure for collecting the structure of the data, this involves the creation of additional tables.

The overall process of normalization for the first three stages is summarized in the figure below.

UN-NORMALIZED DATA

Beware at repeating groups

FIRST NORMAL FORM

If the primary key has more than one field, ensure that all other fields are functionally dependent on the whole key.

SECOND NORMAL FORM

Remove all transitive dependencies i.e depending on non key fields.

THIRD NORMAL FORM

UPDATE ANOMALIES

Update inconsistent data, addition and deletion.

FIRST NORMAL FORMS:- a relation is in first normal form(1F) if it does not contain repeating groups.

Definition: An attribute is a non-key attribute if it is not part of the primary key

SECOND NORMAL FORM: A relation is in second normal form (2F) if it is in first normal form and no non-key attribute is dependent on only a portion of the primary key.

Definition:any attribute (a collection of attributes) that determine any attribute is called a determinant

THIRD NORMAL FORM:A relation is in third normal form (3NF) if it is in second normal form and if the only determinants it contains are candidate keys.

DATABASE MANAGEMENT PRINCIPLES AND APPLICATION

Basic concept of entity-relationship analysis:

What is Entity –Relationship Analysis?

E-R analysis endeavors that:

- Identify major ingredients, entities, of a complicated situation
- Identify characteristics, attributes of the major ingredient
- Uncovers and analyze interconnections, relationships among major entities or ingredient
- Assess various aspects of the relationships

E-R analyses can be summarized as the process of uncovering, analyzing and explicating (i.e stating clearly and fully) the business rule governing a complicated situation

BUSINESS RULES

Rob and cremel say:

- Business rules are precaution statements, derived from detailed description of the organization's operation that defines one or more of the following modern components:
 - o Entities
 - o Relationships

- o Attributes
- o Connectivities
- o Cardinalities
- o Constraints

Examples of business rules:

- An invoice contains one or more invoice lines, but each invoice line is associated with a single invoice.
 - o An invoice contains one or more invoice lines
 - o Each invoice line is associated with a single invoice
 - o A store employs many employees, but each employee is employed by only one store
 - o A college has many departments, but each department belongs to a single college. (this business rule reflects a university that has multiple college such as business, liberal , arts, education, engineering, etc.
 - o A client may sign many contract but each contract is signed by only one client

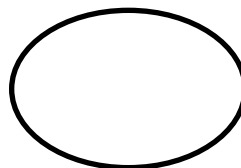
E-R Diagrams

E-R analyses relies heavily on diagrams to represent many components

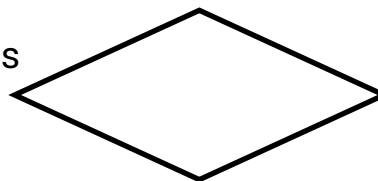
- Entities are represented by rectangles-



- Attributes by ellipses



- Relationships by diamonds



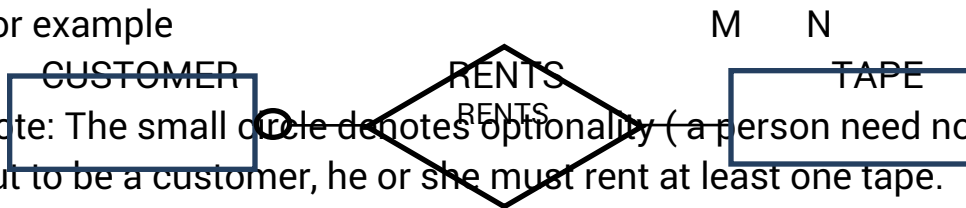
Connections by lines, while _____ may have arrowheads or crow stick to

represent _____ of connectivity or relationship

CRITICAL USE OF E-R ANALYSIS

It is especially important to identify and note many-to-many (M:N) relationships

For example



Note: The small circle denotes optionality (a person need not be a customer) but to be a customer, he or she must rent at least one tape.